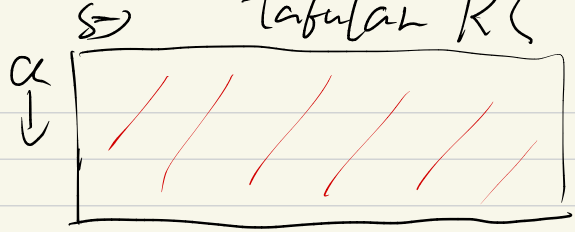
26.03.03

Tabular RL



Value-function Approximation

$V(s, w)$, $Q(s, a, w)$

Gradient Desent to update w.

MC: $\Delta w = \alpha(G_t - V(S, w)) \nabla_w V(S, w)$

unbiased
high variance

Total future reward of the episode.

TD(0) $\Delta w = \alpha(R_t + \gamma V(S_{t+1}, w) - V_t(S_t, w)) \nabla_w V(S_t, w)$

semi-gradient

SARSA: $\Delta w = \alpha(R_t + \gamma \cdot Q(S_{t+1}, A_{t+1}, w) - Q(S_t, A_t, w)) \nabla_w Q(S_t, A_t, w)$

Dueling: $Q(s, a) = A(s, a) + V(s)$

Advantage

Average